

SURYA VENKATA ROHIT MOGANTI

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EDUCATION

State University of New York, Buffalo , Master's in Computer Science and Engineering	Jan 2024 - May 2025
Relevant Coursework: Distributed Computing, Data Intensive Applications, MLOps, LLM Security	
GITAM University, Hyderabad, India , B. Tech in Computer Science and Engineering	Jun 2018 - May 2022
Relevant Coursework: Machine Learning, Algorithm Design, Web Development, Operating Systems	

SKILLS

Programming: Python, SQL, JAVA, Bash
Data Engineering: Pyspark, Kafka, Airflow, Snowflake, Hadoop, Dimensional Modeling, ETL/ELT pipelines
Cloud & DevOps: AWS, Docker, Kubernetes, RESTful APIs, CI/CD (GitHub actions)
Database & Storage: PostgreSQL, DynamoDB, Metastore, ClickHouse
Machine Learning & AI: PyTorch, Hugging Face Transformers, RAG, YOLOv8, Detectron
Additional: Experience with Agile Development, Cursor AI, Streamlit and Next.js

PROFESSIONAL EXPERIENCE

Data Engineer

TEKsystems Global Services	Sept 2022 - Dec 2023
- Built a scalable, event-driven document pipeline using AWS Textract, Lambda, S3, and DynamoDB, improving throughput by 40% and accuracy by 35%. - Developed an integrated human-in-the-loop review system using API Gateway and SageMaker endpoints, enabling real-time manual corrections and logging metadata for audit trails and model retraining. - Enhanced a Next.js frontend for human reviewers and engineered reliable, real-time ingestion workflows with S3 event triggers, CloudWatch, and Python-based fault-tolerant logic, reducing system failures by 90%. - Followed best practices in API design, observability, and scalable architecture to support large-scale data workflows aligned with ELT principles and downstream analytics.	

Junior Data Engineer

TEKsystems Global Services	Feb 2022 - Aug 2022
- Configured and maintained production-grade ETL pipelines across development, test, and production environments using Oracle Data Integrator (ODI 12C), enabling consistent and reliable data integration from diverse source systems into the enterprise data warehouse. - Engineered and optimized data transformation workflows to populate fact and dimension tables, improving performance and scalability for downstream reporting and multi-dimensional analysis.	

Data Engineering Intern

Coromandel International Limited	Jul 2021 - Aug 2021
- Analyzed multi-store sales data to extract purchasing patterns and inventory trends. - Developed a time-series model (Exponential Smoothing) to forecast next purchase dates, enabling stock planning and reducing stockouts	

PROJECTS

E-Commerce Order Insights Pipeline [Link](#)

Aug 2025 - Present

- Orchestrated a modular Airflow ETL pipeline to ingest and transform 100K+ daily e-commerce logs from AWS S3 into PostgreSQL fact/dim tables.
- Implemented data quality checks and automated dashboard updates in Power BI, enabling region-wise customer insights.
- Enabled fault-tolerant DAG execution with alerting and retries, ensuring reliable daily data delivery across dev and prod environments.

Hospital Resource Usage Tracker [Link](#)

Jun 2025 - Jul 2025

- Deployed a real-time data platform to simulate, ingest, and analyze hospital resource logs (50K+ events) across 5 critical departments (ICU, ER, Radiology, etc.), enabling productive resource allocation.
- Engineered an end-to-end PySpark pipeline with Kafka for streaming ingestion and fault-tolerant batch processing, achieving sub-second latency and ensuring 99.9% data availability.
- Designed a Streamlit dashboard with live updates every 5 seconds to visualize utilization trends and detect 85%+ spikes in equipment usage through custom time-series anomaly detection.